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Research

Effect of Electroacupuncture Added to ERAS on Perioperative Parameters and Postoperative Quality of Life in Breast Cancer: A Single-Center, Randomized Controlled Trial

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A B S T R A C T

Purpose: Acupuncture is a potentially beneficial addition to the enhanced recovery after surgery (ERAS) strategy for improving the quality of surgical care for breast cancer. This study evaluated the advantages of acupuncture in postoperative recovery after breast cancer surgery.
Design: A prospective, blinded, randomized, case-control study.
Methods: In this single-center, parallel-group, randomized controlled trial, 144 breast cancer patients undergoing surgery were allocated to the following groups: (group A) conventional group (no acupuncture treatment); (group B) preoperative acupuncture (acupuncture treatment given 1 day before surgery); (group C) intraoperative acupuncture (acupuncture treatment given on the day of surgery); and (group D) a combination of preoperative and intraoperative acupuncture (n = 36/group). The primary outcome was the intraoperative consumption of anesthetics. The secondary outcomes included heart rate and blood pressure changes, intraoperative blood glucose level, pH, and bispectral index, recovery and extubation time, postoperative functional assessment of cancer therapy-breast score, and adverse reactions.
Findings: Intraoperative consumption of sufentanil and blood glucose level was significantly decreased in group C, and no interactive effect was found between the preoperative and intraoperative acupuncture groups. Preoperative heart rate in groups B and C showed significant changes. The 1-week postoperative functional assessment of cancer therapy-breast score was most markedly improved in group C compared with other groups. No adverse reaction occurred with acupuncture.
Conclusions: Intraoperative acupuncture alone is adequate for optimizing the intraoperative state, and preoperative acupuncture seems unnecessary. Acupuncture is safe, with potential benefits for enhanced recovery after surgery in breast cancer surgery.
Trial Registration: Chinese Clinical Trial Registry, ChiCTR1800019979

The incidence of breast cancer has shown a progressive increase over the years.[1,2] Surgery is currently the main treatment for breast cancer. However, surgical outcomes can be affected by the physiological response and measures taken during the perioperative period.[3-5] The concept of enhanced recovery after surgery (ERAS), which was first proposed by a Danish researcher in 1995, involves the application of evidence-based multimodal therapies in the perioperative period to alleviate the patient's physiological stress response. This approach has been widely adopted in many surgical disciplines.[6]

Breast cancer is among the most common cancers in women.[7] Thus, the development of a relevant model for ERAS in patients with breast cancer is a key imperative. Acupuncture has been demonstrated to enhance perioperative rehabilitation of patients[8] and

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improve perioperative care quality.[9] The addition of acupuncture in ERAS may provide new methods and ideas for promoting recovery.[10]

The concept of "perioperative acupuncture medicine" is aimed to establish an optimal therapeutic approach in the perioperative period for surgical patients. This includes the application of acupuncture as part of preoperative, intraoperative, and postoperative management.[8] Use of acupuncture in anesthesia has been shown to reduce the amount of anesthetics used, which can help maintain hemodynamic stability. Acupuncture can optimize the patient's preoperative condition, stabilize blood pressure and blood sugar level, reduce the risk of intraoperative anesthesia and postoperative complications, and accelerate patient rehabilitation.[8]

In this study, we assessed the use of acupuncture with ERAS in breast cancer surgery and compared the advantages of its application at different timepoints in the perioperative period. Comparison was conducted between the conventional method and acupuncture given 1 day before surgery (preoperative), on the day of surgery (intraoperative), and a combination of both preoperative and intraoperative acupuncture. No previous study has investigated the effect of such combination intervention, and thus it is necessary to explore its significance. The aim of this study was to develop an optimal acupuncture timing for optimizing the perioperative state and recovery after breast cancer surgery. We explored the effect of preoperative and intraoperative acupuncture on the patient, and assessed whether these interventions need to be performed concurrently to produce a better effect, or whether just one-time acupuncture can be effective.

Methods

Ethical Approval

This study was granted approval by the Ethics Committee of the Shuguang Hospital (no. 2018-624-53-01) and was conducted in accordance with the 1964 Helsinki Declaration or comparable standards. All participants provided their written informed consent before enrollment (trial registration number: ChiCTR1800019979).

Clinical Data

Between November 2018 and December 2019, 144 patients who were admitted for breast-conserving surgery with a radical mastectomy were enrolled. The operating procedures were performed by the same team of surgeons and anesthesiologists, and strictly conformed to the standard guidelines. The flowchart of the study is presented in Figure 1.

Inclusion criteria: (1) breast cancer patients who conform with the diagnostic criteria of Traditional Chinese Medicine[11] and Western medicine,[12] tumor, nodes, and metastases stage I to III, and no signs of systemic metastasis; (2) age 16 to 70 years; (3) American Association of Anesthesiologists Grade I and II[13]; (4) completed all auxiliary examinations for admission; (5) no history of immune system diseases.

Exclusion criteria: (1) patients who had malignant hypertension, diabetes, or severe heart disease; (2) symptoms of insomnia in the preceding 6 months; (3) taking sleeping pills in the preceding 6 months; (4) patients with coagulation dysfunction, (5) pregnant or lactating women; (6) previous history of anxiety or depression; (7) patients who had mental illness or dementia; (8) local infection at acupoints; (9) participated in other clinical trials within 4 weeks.

Study Design

SPSS24.0 (IBM Corp.) software was used for random grouping, and based on this, the experimental methods were developed and assigned to corresponding grouped patients, thus ensuring balance among the different groups. A person was specially assigned to perform the randomization grouping work of SPSS24.0 software, accurately recorded, and determined the "random allocation table" with grouping information. The results of grouping randomization were made into cards, numbered, and placed in a sealed opaque envelope for group allocation. Q.T. generated the random allocation sequence, R.L. enrolled participants, and Y.G. assigned participants to interventions. The patients and investigators performing the data collection and analysis were blinded to the experimental grouping.

Experimental Grouping

A total of 144 patients were randomly allocated to (A) conventional, (B) preoperative acupuncture, (C) intraoperative acupuncture, and (D) a combination of preoperative and intraoperative acupuncture groups (n = 36/group).

Experimental Procedure

Before Operation

One day before surgery, the baseline information, which included body mass index, heart rate, and blood pressure, was collected. Electrocardiogram monitoring was done.

Day of Operation

After establishing an intravenous line, sodium lactate Ringer's solution was administered to the patient. Monitoring of pulse oxygen saturation, electrocardiogram, and blood pressure (mean arterial pressure) was performed. Total intravenous anesthesia was used to assist tracheal intubation. Propofol 1 to 2 mg/kg, sufentanil 0.3 mcg/kg, and rocuronium 0.6 mg/kg were used for anesthesia induction. During surgery, propofol 2 to 3.5 mcg/mL and remifentanil 0.05 to 0.2 mcg/kg/min were administered via the target-controlled injection pump. Sufentanil was added if necessary, and 30 to 40 minutes after surgery, 1/5 of the initial dose of rocuronium was administered. The goal of drug dose adjustment was to produce a bispectral index (BIS) reaching 40 to 60 stimuli with no body movement response. The dosages of remifentanil and sufentanil were modified in case of significant changes in heart rate and blood pressure during skin incision. If the BIS did not reach the standard value, the propofol concentration was adjusted. The anesthetic administration was based on the patient's body weight, and there were no differences in body weight among the 4 groups of patients. After completion of surgery, the patients were extubated.

Group A

No acupuncture was given.

Group B

The patients received acupuncture 1 day before surgery. Acupoints were selected based on the recommended guideline[14] and national standard of the People's Republic of China GB 12346-90 Location of points.

After needle insertion, reinforcing-reducing technique was used at Shenmen (HT 7), Neiguan (PC 6), Yintang (EX-HN3), and Baihui (DU 20) acupoints. The sterilized disposable needles were inserted using both hands under aseptic conditions. A G6805-2 therapy device was connected to Yintang-Baihui and Shenmen-Neiguan. The frequency was set to 2 Hz, continuous wave, and the current intensity was adjusted according to the patient's tolerance to needle sensation. The treatment was performed continuously for 30 minutes.

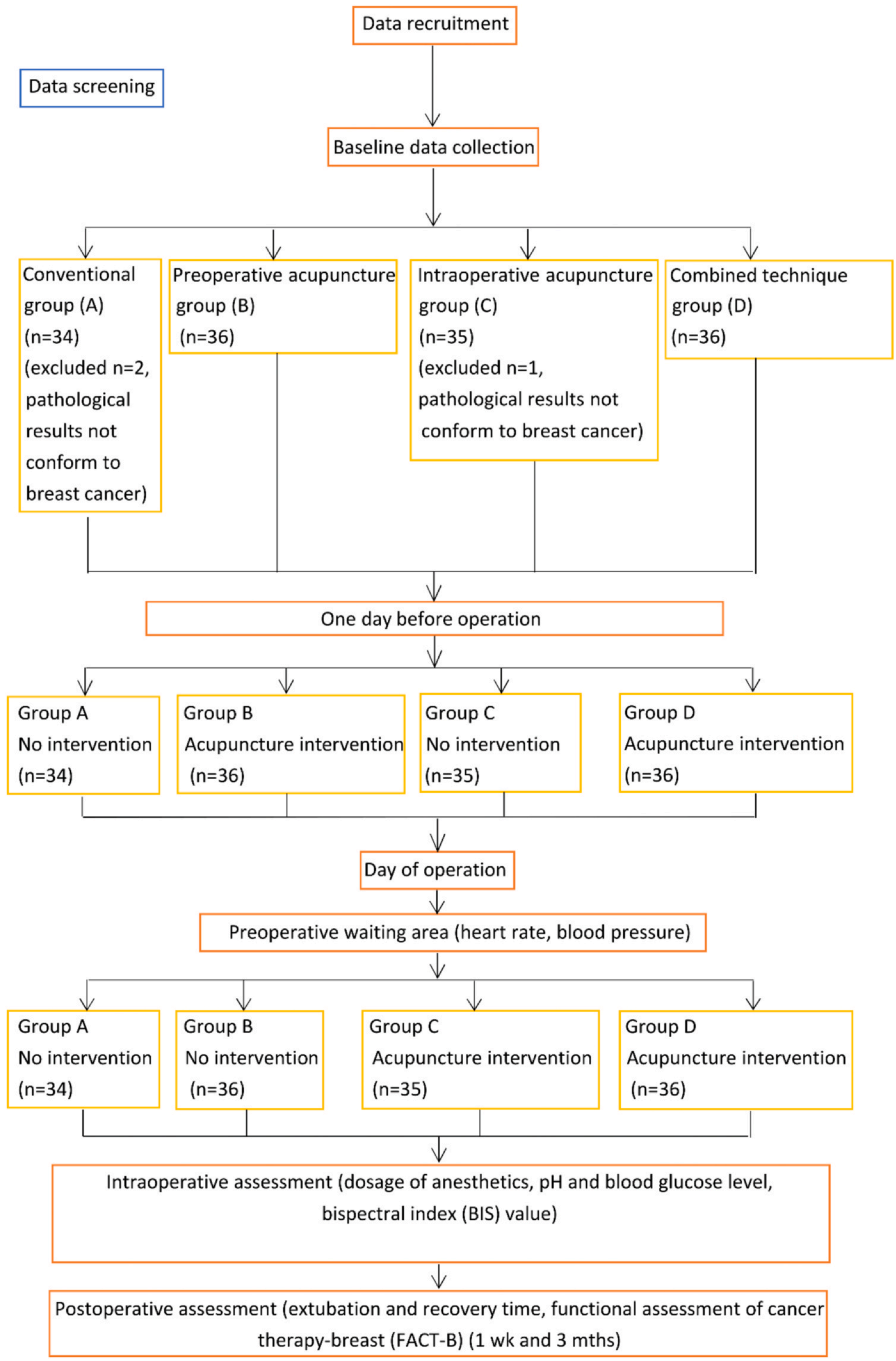

Figure 1. Flowchart of the study. This figure is available in color online at www.jopan.org.

Group C

The patients received acupuncture on the day of operation. In the preoperative waiting area, after needle insertion, reinforcing-reducing technique was used at bilateral Hegu (LI 4), Neiguan (PC 6), and Liangqiu (ST 34). The sterilized disposable needles were inserted using both hands under aseptic conditions. The G6805-2 therapy device was connected to bilateral Hegu, Neiguan, and Liangqiu (at 1 inch outward). The frequency was set to 2 Hz, continuous wave, and the current intensity was adjusted according to the patient's tolerance to needle sensation. The treatment was performed continuously for 30 minutes. Additionally, Yintang and Baihui acupoints were used to enhance the sedative effect, and treatment was performed for more than 30 minutes. The electroacupuncture (EA) was continued in the operating room. The needles were removed once the patient was awake.

Group D

The patients received acupuncture and EA 1 day before as well as on the day of operation as in groups B and C, respectively.

Observation Indicator

Primary outcome: intraoperative anesthetics consumption (total amount of anesthetics given).

Secondary outcomes: heart rate and blood pressure changes, intraoperative pH and blood glucose level, BIS, extubation and recovery time, and functional assessment of cancer therapy-breast (FACT-B) at postoperative 1 week and 3 months (by telephonic follow-up).

Data Analysis

SPSS24.0 was used for data analysis. Patients who completed the entire experiment were included for analysis. The continuous variables were presented as mean ± standard deviation and between-group differences assessed using the *t* test. Analysis of variance (ANOVA) with least-significance difference and Student-Newman-Keuls posthoc tests were used for multigroup comparisons. χ^2 test was used for comparison of categorical variables. ANOVA of factorial design was used to compare the main observation indicators of patients before and during operation, and the interaction between the 2 factors was analyzed.

Sample Size Calculation

The main indicator was the difference in anesthesia consumption between the acupuncture and the conventional (general anesthesia) groups. A preliminary study was performed to assess the feasibility of the study and select the appropriate sample size. The amount of sufentanil anesthesia used in the control group and acupuncture group (n = 10 per group) was calculated to be 44.62 ± 3.00 mcg and 42.20 ± 3.00 mcg, respectively. Assuming $\alpha = 0.05$, power $1 - \beta = 0.90$, and a potential 10% dropout factor, the sample size was determined using the following formula:

$$n = 2\,(Z\alpha + Z\,[1 - \beta])^2 \times SD^2/d^2$$

where Z is the constant and d is the effect size,

The adjusted sample size N1 = n/(1 − dropout rate) = 36 patients per group.[15]

Clinical Safety Evaluation

The clinical safety was assessed based on patient self-report or daily observation by the physician and during follow-up. Any acupuncture side effects were managed strictly according to the recommended guidelines.[16] The adverse events were classified as mild, moderate, and severe. The correlation of the occurrence of adverse events during the study with the design of the intervention can be classified into the following categories: unable to judge; definitely unrelated; possibly unrelated; possibly related; and definitely related.

Results

Baseline Data Comparison

The age, body mass index, operation time, preoperative blood glucose, heart rate, and blood pressure were not significantly different between the 4 groups ($P > .05$) (Table 1). There were 3 dropout cases in which the postoperative pathological findings did not conform to breast cancer (group A, 2 cases; group C, 1 case).

Primary Outcome

Intraoperative Anesthetics Consumption

As shown in Table 2a, the amount of sufentanil used was significantly different among the 4 groups ($P < .05$); however, there was no significant difference with respect to the consumption of propofol, rocuronium, and remifentanil ($P > .05$). The amount of sufentanil used in groups D and C was significantly less than that in group A ($P < .01$ and $P < .05$, respectively), supporting that acupuncture intervention can reduce the intraoperative consumption of sufentanil.

Further analysis revealed significant difference in the amount of sufentanil used in group C ($P < .05$), but no significant differences in groups B and D ($P > .05$). Intraoperative acupuncture had an effect on the amount of sufentanil used, and there was no interaction between the preoperative and intraoperative acupuncture groups (Table 2b).

Secondary Outcomes

Heart Rate and Blood Pressure

As shown in Table 3a, in the preoperative waiting area, there was a marked difference in the heart rate ($P < .01$) in each group; however, there were no significant differences in the blood pressure ($P > .05$). The heart rate in group D was significantly lower than that in group A ($P < .01$).

We further evaluated any potential interaction between the preoperative and intraoperative acupuncture interventions (Table 3b). The heart rate changes in groups B and C exhibited significant differences ($P < .05$), while group D showed no significant difference ($P > .05$). Both preoperative and intraoperative

Table 1
Baseline Parameters in Each Group

Baseline Parameter	Conventional Group (n = 34)	Preoperative Acupuncture (n = 36)	Intraoperative Acupuncture (n = 35)	Combined Technique Acupuncture (n = 36)	*F*	*P*
Age (y)	53.32 ± 10.78	51.92 ± 10.69	53.51 ± 10.85	52.14 ± 12.11	0.19	.90
Operation duration (min)	130.12 ± 10.04	128.36 ± 6.76	128.43 ± 8.15	126.61 ± 8.29	1.03	.38
BMI	20.93 ± 2.01	21.71 ± 2.49	22.06 ± 2.42	21.44 ± 1.76	1.64	.18
Preoperative blood glucose (mmol)	5.52 ± 0.98	5.30 ± 0.96	5.01 ± 0.73	5.49 ± 1.14	2.14	.10
Heart rate (bpm)	80.94 ± 4.31	82.39 ± 3.94	82.37 ± 4.39	81.31 ± 4.55	1.04	.38
Systolic blood pressure (mm Hg)	129.53 ± 6.03	129.47 ± 5.86	128.49 ± 5.63	128.92 ± 6.02	0.25	.86
Diastolic blood pressure (mm Hg)	77.41 ± 4.02	77.36 ± 3.97	77.37 ± 4.85	75.64 ± 4.25	1.48	.22

BMI, body mass index.
Data presented as mean ± standard deviation.

Table 2a
Anesthesia Consumption in Each Group

Group	Number of Cases	Propofol (mg)	Rocuronium (mg)	Sufentanil (mcg)	Remifentanil (mg)
Conventional (group A)	34	1006.12 ± 71.62	48.32 ± 2.41	44.53 ± 3.48	0.72 ± 0.06
Preoperative acupuncture (group B)	36	1009.72 ± 67.42	48.50 ± 2.43	43.39 ± 3.74	0.72 ± 0.07
Intraoperative acupuncture (group C)	35	986.71 ± 74.11	48.40 ± 2.52	42.57 ± 3.38*	0.68 ± 0.10
Combined technique acupuncture (group D)	36	978.28 ± 72.81	47.64 ± 2.05	42.17 ± 3.58†	0.72 ± 0.13
F		1.60	0.99	3.00	1.30
P		.19	.40	.03	.28

Data presented as mean ± standard deviation.
* $P < .05$ versus group A.
† $P < .01$ versus group A.

Table 2b
Analysis of Consumption of Anesthetics in Each Group

Source of Variation	Degree of Freedom	SS	MS	*F*	*P*
Total variation	141	264,356.00			
Preoperative acupuncture (group B)	1	21.03	21.03	1.67	.20
Intraoperative acupuncture (group C)	1	89.08	89.08	7.08	.01
Combined technique acupuncture (group D)	1	4.77	4.77	0.38	.54
Error	137	1724.60	12.59		

SS, sum of squares; MS, mean square
Data were analyzed with analysis of variance of factorial design.

Table 3a
Heart Rate and Blood Pressure in the Preoperative Waiting Area

Group	Number of Cases	Heart Rate (bpm)	Systolic Pressure (mm Hg)	Diastolic Pressure (mm Hg)
Conventional (group A)	34	76.44 ± 6.34	125.24 ± 11.94	73.32 ± 9.01
Preoperative acupuncture (group B)	36	73.67 ± 8.65	125.75 ± 14.95	73.97 ± 9.24
Intraoperative acupuncture (group C)	35	73.54 ± 6.42	123.54 ± 13.07	70.97 ± 8.35
Combined technique acupuncture (group D)	36	71.00 ± 8.29*	125.72 ± 11.18	73.97 ± 7.95
F		3.05	0.23	0.96
P		.03	.88	.42

Data presented as mean ± standard deviation.
* $P < .01$ versus group A.

Table 3b
Analysis of the Heart Rate in the Preoperative Waiting Area in Each Group

Source of Variation	Degree of Freedom	SS	MS	*F*	*P*
Total variation	141	772,561.00			
Preoperative acupuncture (group B)	1	249.03	249.03	4.40	.04
Intraoperative acupuncture (group C)	1	272.76	272.76	4.82	.03
Combined technique acupuncture (group D)	1	0.47	0.47	0.01	.93
Total variation	137	7751.07	56.58		

SS, sum of squares; MS, mean square.
Data were analyzed with analysis of variance of factorial design.

acupuncture had a positive effect in stabilizing the heart rate, but the effects of the two were independent of each other.

Intraoperative Blood Glucose and pH Level

As shown in Table 4a, the intraoperative blood glucose level was substantially different among the 4 groups ($P < .05$), but the pH values were not significantly different ($P > .05$). The fluctuation in blood glucose level in group D was significantly different compared with groups A and B ($P < .05$), while that in group C was significantly different compared with group A ($P < .05$).

Further analysis showed marked changes in the intraoperative blood glucose level in group C ($P < .01$), but no significant changes in groups B and D ($P > .05$). Intraoperative acupuncture reduced intraoperative blood glucose level, and there was no interaction between the preoperative and intraoperative acupuncture groups (Table 4b).

Table 4a
Blood Glucose and pH Value in Each Group

Index	Conventional (Group A) (n = 34)	Preoperative Acupuncture (Group B) (n = 36)	Intraoperative Acupuncture (Group C) (n = 35)	Combined Technique Acupuncture (Group D) (n = 36)	*F*	*P*
Blood glucose (mmol)	6.17 ± 0.40	6.07 ± 0.67	5.83 ± 0.98*	5.75 ± 0.48*	3.03	.03
pH value	7.39 ± 0.03	7.40 ± 0.02	7.39 ± 0.02	7.40 ± 0.03	0.61	.61

Data presented as mean ± standard deviation.
* $P < .05$ versus group A.

Table 4b
Analysis of Blood Glucose Level in Each Group

Source of Variation	Degree of Freedom	SS	MS	*F*	*P*
Total variation	141	5062.66			
Preoperative acupuncture (group B)	1	0.28	0.28	0.62	.43
Intraoperative acupuncture (group C)	1	3.82	3.82	8.52	< .01
Combined technique acupuncture (group D)	1	0.01	0.01	0.01	.91
Error	137	61.47	0.45		

SS, sum of squares; MS, mean square.
Data were analyzed with analysis of variance of factorial design.

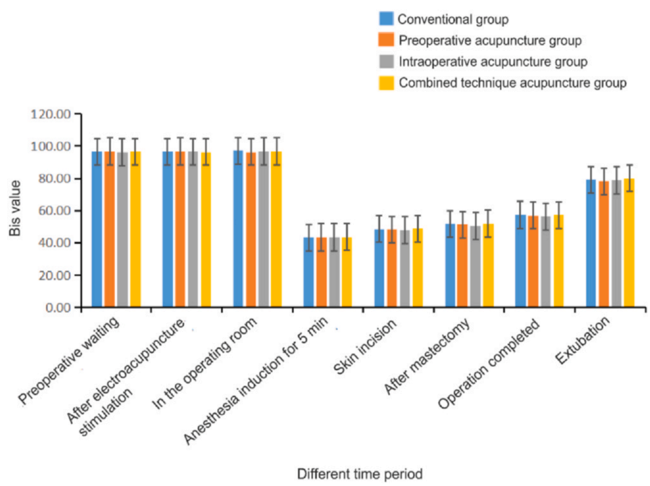

Figure 2. BIS value changes in each time period. BIS, bispectral index. This figure is available in color online at www.jopan.org.

BIS Value Changes

As shown in Figure 2, the BIS value of the 4 groups showed no significant difference for each time period ($P > .05$).

Extubation and Recovery Time

As shown in Table 5, there were no significant differences in the extubation or recovery time among the 4 groups ($P > .05$).

Postoperative FACT-B Score at 1 Week and 3 Months

As shown in Table 6a, ANOVA analysis demonstrated that the FACT-B score at 1 week postoperatively was remarkably different among the 4 groups ($P < .05$), but not at 3 months postoperatively ($P > .05$).

We conducted ANOVA of factorial design for the significant results obtained at 1 week postoperatively (Table 6b). The FACT-B score in group C was markedly different ($P < .05$), while there was no significant difference in groups B and D ($P > .05$). The results suggest that intraoperative acupuncture improved the FACT-B score at 1 week postoperatively and the quality of life of patients. There was no interaction between the preoperative and intraoperative acupuncture groups.

Table 6a
FACT-B in Each Group

Group	Number of Cases	Week 1	3 Months
Conventional (group A)	34	89.32 ± 2.96	109.29 ± 5.92
Preoperative acupuncture (group B)	36	89.53 ± 2.85	109.92 ± 5.25
Intraoperative acupuncture (group C)	35	89.77 ± 2.89	109.40 ± 6.32
Combined technique acupuncture (group D)	36	91.33 ± 3.98	110.22 ± 5.03
F		2.89	0.21
P		.04	.89

ANOVA, analysis of variance; FACT-B, functional assessment of cancer therapy-breast.
Data were analyzed with ANOVA. Data presented as mean ± standard deviation.

Adverse Reactions

No adverse events occurred during the study period.

Discussion

Acupuncture and Its Benefits

Acupuncture has been demonstrated to promote the perioperative rehabilitation of patients. Moreover, there have been rapid developments in acupuncture methods. Acupuncture has been shown to relieve stress in the preoperative stage, maintain homeostasis during surgery, enhance recovery, and alleviate postoperative pain.[8]

The perioperative period has a great impact on the patient. From the moment the decision for surgery is made, the patient may be in a state of worry about the whole process of the operation. On the day before the operation, a nurse routinely instructs the patient about the preoperative precautions, and the anesthesiologist also evaluates the patient 1 day before the operation. The conversation is mainly about some preoperative precautions and anesthesia methods, including the potential complications. Therefore, the acupuncture and moxibustion team can also visit the patients on the day before the operation and provide assistance. Particularly on the day of the operation, during the preoperative waiting period, as the surgeon and nurses in the operating room are busy with a series of preoperative preparations, the anxiety and fear of the patient who is lying on the operating table tends to be ignored. The nurses may provide some psychological comfort, but sometimes this help is not enough. Acupuncture intervention may fill this gap, because it may help relieve the preoperative tension and anxiety and help maintain hemodynamic stability, so that the operation can be performed more smoothly.[17]

Table 5
Extubation and Recovery Time in Each Group

Indicator (min)	Conventional (Group A) (n = 34)	Preoperative Acupuncture (Group B) (n = 36)	Intraoperative Acupuncture (Group C) (n = 35)	Combined Technique Acupuncture (Group D) (n = 36)	*F*	*P*
Extubation time	10.32 ± 1.61	9.83 ± 1.83	9.89 ± 1.62	9.58 ± 1.50	1.22	.31
Recovery time	27.65 ± 1.84	27.69 ± 1.85	27.49 ± 2.02	27.11 ± 1.91	0.69	.56

Data presented as mean ± standard deviation.

Table 6b
Analysis of FACT-B Score 1 Week Postoperatively in Each Group

Source of Variation	Degree of Freedom	SS	MS	*F*	*P*
Total variation	141	1,143,602.00			
Preoperative acupuncture (group B)	1	27.47	27.47	2.67	.11
Intraoperative acupuncture (group C)	1	44.73	44.73	4.34	.04
Combined technique acupuncture (group D)	1	16.23	16.23	1.57	.21
Error	137	1,412.59	10.31		

ANOVA, analysis of variance; FACT-B, functional assessment of cancer therapy-breast.
Data were analyzed with ANOVA of factorial design.

In this study, we explored the effect of acupuncture performed 1 day before the operation and the effect of acupuncture on the day of operation (at the preoperative waiting area and intraoperatively). The acupoints selected for group B preoperative acupuncture were for the purpose of alleviating preoperative anxiety, while for group C intraoperative acupuncture, the acupoints were selected to reduce intraoperative pain and maintain intraoperative hemodynamic stability. Group D was a combination of both. As the effect of such multiple interventions has not been investigated in previous studies, it is necessary to explore its significance.

Effect of Electroacupuncture on Breast Cancer Surgery Patients

Intraoperative Circulation and Anesthesia

Acupuncture reduced intraoperative anesthetics consumption, and thus reduced the possibilities of the associated adverse reactions. In a previous study, acupuncture was found to decrease the need for anesthetic use by 10% to 30%, and in some cases by up to 50%.[18] EA at the lateral Neimadian and Neiguan 30 minutes before and after anesthesia in esophageal cancer surgery was shown to significantly reduce the intraoperative dose of sulfentanil.[19] Transcutaneous electrical acupuncture stimulation (TEAS) reduced the intraoperative opioid dosage in patients undergoing video-assisted thoracoscopic surgery lobectomy, and the consumption was found to be the lowest at 2/100 Hz.[20] In the present study, intraoperative acupuncture significantly reduced the amount of sufentanil used, while the intraoperative heart rate and blood pressure remained relatively stable. Sulfentanil is a synthetic opioid that is widely used for analgesia and anesthesia.[21,22] This indicates that acupuncture has a certain analgesic effect, and it may possibly enhance the effect of conventional analgesics. Previous research suggested that EA induces an increase in the binding affinity and density of opioid receptors, and a combination of acupuncture with drugs may further enhance the EA effect.[23] Acupuncture can be used to obtain or augment anesthesia by localizing acupuncture points according to diseases and surgical sites.[24] Combining acupuncture with anesthesia has been shown to improve clinical outcomes, including reduced preoperative anxiety, lowered intraoperative stress response, improved immune function, and fewer postoperative side effects.[25-29] Acupuncture can not only reduce the use of anesthetics and analgesics, but also the related complications, and improve organ function in the perioperative period.[30] Rational application of acupuncture intraoperatively can help in stabilizing the respiratory and circulatory functions, as well as in alleviating surgical and anesthetic stress.[8] TEAS at 2/100 Hz has also been shown to reduce intraoperative remifentanil consumption[31]; however, we did not observe any significant change in this study probably due to the difference in the technique used (such as stimulation frequency).

Preoperative Heart Rate Stability

Previous studies have demonstrated that acupuncture has a distinct effect on the central and the autonomic nervous system, which can regulate the heart rate variability.[32-34] The ability of acupuncture in improving heart rate variability can be used as a tool to monitor treatment effectiveness and the impact on the quality of life of patients.[34] In this study, both preoperative and intraoperative acupuncture showed a positive effect in stabilizing the heart rate, and the effects of the two groups were independent. The heart rate in the preoperative waiting area may reflect the state of anxiety and stress before surgery. Acupuncture may help optimize the preoperative state and reduce anxiety as the heart rate is stabilized. It shows that both preoperative acupuncture and intraoperative acupuncture have positive effects on the stabilization of heart rate, but that there is no interaction between the two groups. This suggests that acupuncture on the day before surgery or on the day of surgery can stabilize the heart rate in the preoperative waiting area. Therefore, there is no need for simultaneous acupuncture, which can help reduce unnecessary interventions.

In response to stress, activation of the sympathetic-adrenomedullary and hypothalamic-pituitary-adrenocortical axes may increase the levels of adrenaline (epinephrine), noradrenaline (norepinephrine), and cortisol, which may then increase the pulse rate, blood pressure, and blood glucose levels to cope with stress. Previous research suggested that EA may play an important protective role by acting through these axes.[35,36] Acupuncture may decrease the heart rate in humans through cardiac vagal activity stimulation and cardiac suppression of the sympathetic activity.[35]

Intraoperative Blood Glucose Level

In this study, intraoperative acupuncture was found to reduce intraoperative blood sugar level. Intraoperative change in blood glucose level is one of the important indicators for monitoring surgical stress response. In some special operations, intraoperative stress hyperglycemia was shown to be associated with significantly poor long-term prognosis.[37] Therefore, effective control of intraoperative blood sugar level may also improve the prognosis of such patients.[38] TEAS was shown to reduce the intracoronary stress index levels, including blood glucose during craniotomy.[39] In patients undergoing modified radical mastectomy for breast cancer, EA stabilized the intraoperative hemodynamics and blood sugar levels and reduced the stress response.[40] Previous research suggested that EA could stimulate an increase in the whole-body glucose uptake mediated by activation of the autonomic nervous system.[41]

Postoperative Quality of Life

Previous studies revealed that acupuncture may improve the quality of life of patients undergoing breast cancer treatment.[42] Our study suggested that intraoperative acupuncture markedly improved the FACT-B score 1 week postoperatively and the quality of life. Previous studies have found that acupuncture-induced analgesia may have an organ-protective effect,[30] which might contribute to the improved postoperative quality of life. Acupuncture enhances postoperative analgesia,[30] and has been shown to improve hemodynamic stability and cardiac outcome.[43] It can protect the lungs against perioperative insult and improve postoperative lung function, which is probably attributable to the decreased use of anesthetics or a direct mechanism.[30,44] During surgery, several factors may affect the blood flow in liver and kidneys, while acupuncture can relieve the adverse effects of these pathophysiological changes.[45]

Application of ERAS in Breast Cancer

ERAS is a collective, standardized, and evidence-based set of preoperative, intraoperative, and postoperative multidisciplinary interventions to improve perioperative care. ERAS has been validated in many surgical specialties, including colorectal surgery, urology, and gynecology.[46,47] ERAS can alleviate the financial burden on patients and their families, as well as hospitals and the society.[48]

Breast cancer is common among women in the age group of 40 to 60 years, and its incidence rate has shown a progressive increase over the years.[49,50] Currently, the two main surgical treatment methods for early-stage breast cancer are modified radical mastectomy and breast-conserving surgery. Studies have shown that the latter method has better therapeutic effect.[51] Breast-conserving surgery is the most commonly performed breast cancer surgery in our hospital, therefore, we selected this method for our study.

In patients with breast cancer, ERAS approach has been applied for mastectomy and breast-reconstructive surgery.[52,53] Its implementation aims to hasten postoperative recovery.[53] Previous studies have demonstrated that ERAS contributes to accelerated postoperative recovery of patients undergoing deep inferior epigastric perforator flap breast reconstruction.[53,54]

Limitations

We did not include a sham acupuncture group in this study. This is because, in general, the main types of auxiliary tools used for sham acupuncture are blunt needles, ordinary acupuncture needles, TEAS instruments, or other nonacupuncture placebo needles, and the effect of sham acupuncture may include the specific and nonspecific effects of acupuncture in addition to its placebo effect that may interfere with the results.[55] In previous studies, acupuncture was shown to shorten the postoperative extubation and recovery time.[31,56] Moreover, EA was also shown to lower the BIS value in patients undergoing colonoscopy.[57] However, we did not observe the changes in this study probably due to the limitation of sample size. Besides, addition of specific acupoints according to the specific needs of individual patients may help enhance the effect. Future studies may involve multicenter studies with larger sample size. In addition, the long-term benefits of acupuncture should also be explored.

Conclusion

In this study, intraoperative acupuncture was found to reduce the intraoperative use of sufentanil. In addition, it also decreased the intraoperative blood glucose level. Of note, there was no interactive effect between the preoperative and intraoperative acupuncture groups. Both preoperative and intraoperative acupuncture improved the preoperative heart rate in the preoperative waiting area. Intraoperative acupuncture most significantly improved the 1-week postoperative FACT-B score in patients. Therefore, for optimizing the intraoperative state, intraoperative acupuncture alone may be adequate, which can reduce unnecessary use of preoperative acupuncture in the future. Acupuncture was found to be safe, and its addition in ERAS may help enhance the postoperative recovery and outcomes of breast cancer surgery.

Declaration of Competing Interest

None to report.

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